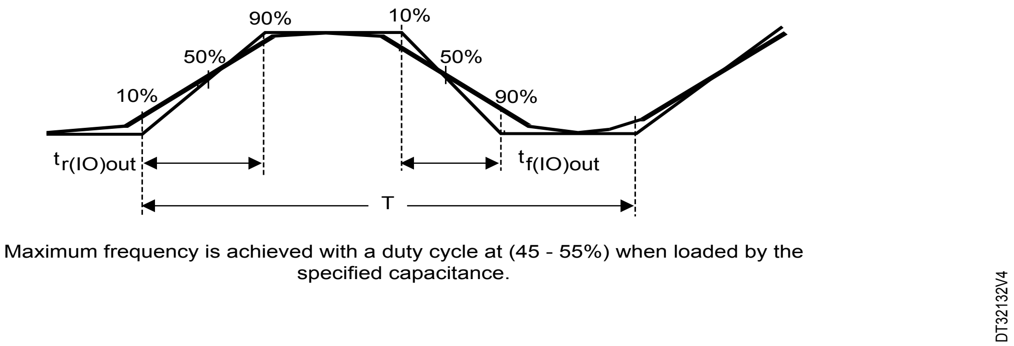
Figure 21. Output AC characteristics definition



5.3.15 NRST pin characteristics

The NRST pin input driver uses the CMOS technology. It is connected to a permanent pullup resistor, R_{PU}.

Unless otherwise specified, the parameters given in the table below are derived from tests performed under the ambient temperature and supply voltage conditions summarized in Table 16.

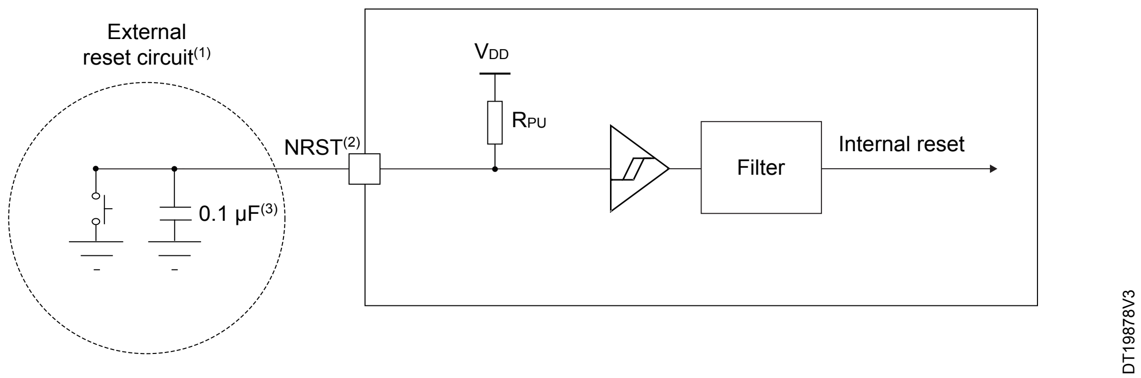
Table 50. NRST pin characteristics

Specified by design and not tested in production.

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{IL(NRST)}$	NRST input low-level voltage	-	-	-	0.3 x V_{DDIOx}	V
$V_{IH(NRST)}$	NRST input high-level voltage	-	0.7 x V_{DDIOx}	-	-	
$V_{hys(NRST)}$	NRST Schmitt trigger voltage hysteresis	-	-	200	-	mV
R_{PU}	Weak pull-up equivalent resistor(1)	$V_{IN} = V_{SS}$	30	40	50	kΩ
$t_{F(NRST)}$	NRST input filtered pulse	-	-	-	50	ns
$t_{NF(NRST)}$	NRST input not-filtered pulse	2.7 V ≤ V_{DD} ≤ 3.6 V	350	-	-	

1. *The pull-up is designed with a true resistance in series with a switchable PMOS. This PMOS contribution to the series resistance is minimal (~10 % order).*

Figure 22. Recommended NRST pin protection



(1): The reset network protects the device against parasitic resets.

(2): The user must ensure that the level on the NRST pin can go below the $V_{IL(NRST)}$ max level specified in the above table. Otherwise the reset is not taken into account by the device.

(3): The external capacitor on NRST must be placed as close as possible to the device.